E. Mahima - 192312227

ITA0329 - MOBILE COMPUTING CO1-AT-1 CONCEPT MAPPING

5) A =

Mobile Network protocol stack

Goal
Reliable, Secure and efficient mobile communication

CONTRIBUTION

Contribution
- provide network services
- Handles data formatting & encryption

INTERACTION

- Uses transport layer services
- Works with lower layers for data delivery

Application Layer (User Services)

Data (up/down)

- Ensures reliable delivery
- Flow control & error recovery
- (TCP/UDP)

Transport Layer (End-to-End)

- Receives data from application layer

Segments/packets

- Logical addressing & routing
- Mobility management
- Reaches correct Nlw / device

Network Layer (Routing & Mobility)

- Uses transport layer data
- provides datagrams to data link layer

frames

- Framing, error detection/correction
- Medium access control (MAC)

Data link layer (Framing & Error Control)

- Receives datagrams from Nlw layer
- sends frames to physical layer

Bits

- Converts data to radio signals
- Modulation, coding, Spectrum
- Reliable wireless transmission

physical Layer (Radio transmission)

- Receives frames from data link layer
- Transmits signals over the air